Pb 1. K of $\mathbb{E}^2$

$A(-5,2)$, $B(4,-1)$, $C(3,6) \to \Delta ABC$.

K' coord. sys. w origin A and basis $\mathcal{B}' = (\vec{AB}, \vec{AC})$.

a) ΔABC isosceles?

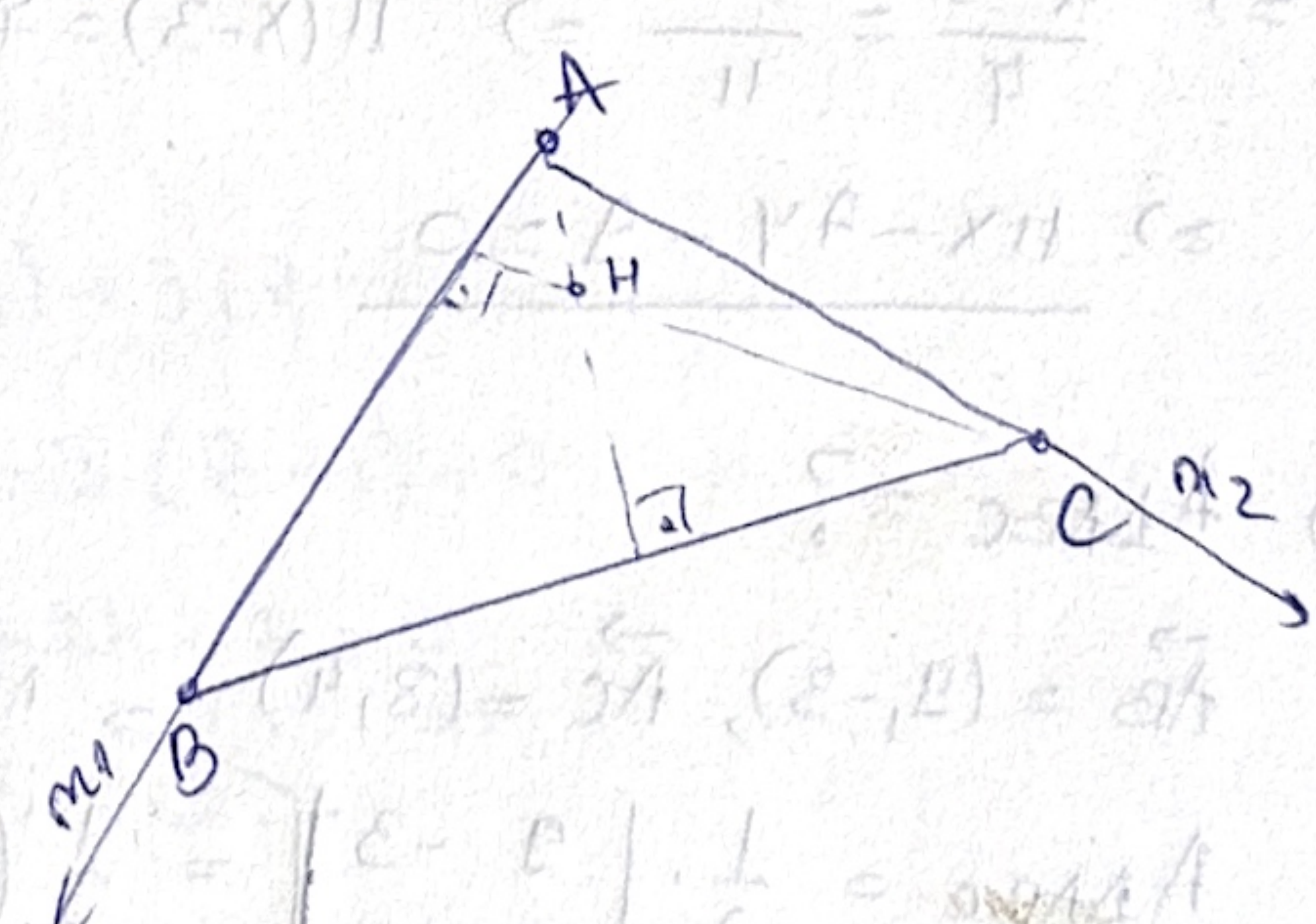
$\vec{AB} = (9,-3) \Rightarrow |\vec{AB}| = 9^2+(-3)^2 = 81+9 = 90.$

$\vec{AC} = (8,4) \Rightarrow |\vec{AC}| = 8^2+4^2 = 64+16 = 80$ $\Big| \Rightarrow \Delta$ NOT is.

$\vec{BC} = (-1,7) \Rightarrow |\vec{BC}| = (-1)^2+7^2 = 1+49 = 50.$

b) K → orthocenter of ΔABC.

Let $H(x,y) \to \cap \perp$.



$AH \perp BC$

$\vec{BC}(-1,7)$

Normal form of a line:

$\boxed{\text{Line: } \langle m, X-A \rangle = 0.}$

$\langle \vec{AB}, (x-x_C, y-y_C) \rangle$

$\vec{AB} = (9,-3) \Rightarrow m_1 = \vec{AB}$

$\langle (9,-3), (x-3, y-6) \rangle = 0$

$\Rightarrow 9(x-3) - 3(y-6) = 0$

$\Rightarrow 9x - 3y - 9 = 0 \Rightarrow \underline{3x - y - 3 = 0}$

$\vec{AC} = (8,4) \Rightarrow m_2 = \vec{AC}$

$\langle (8,4), (x-4, y+1) \rangle = 0$

$\Rightarrow 8(x-4) + 4(y+1) = 0$

$\Rightarrow 8x + 4y - 28 = 0 \Rightarrow \underline{2x + y - 7 = 0}$

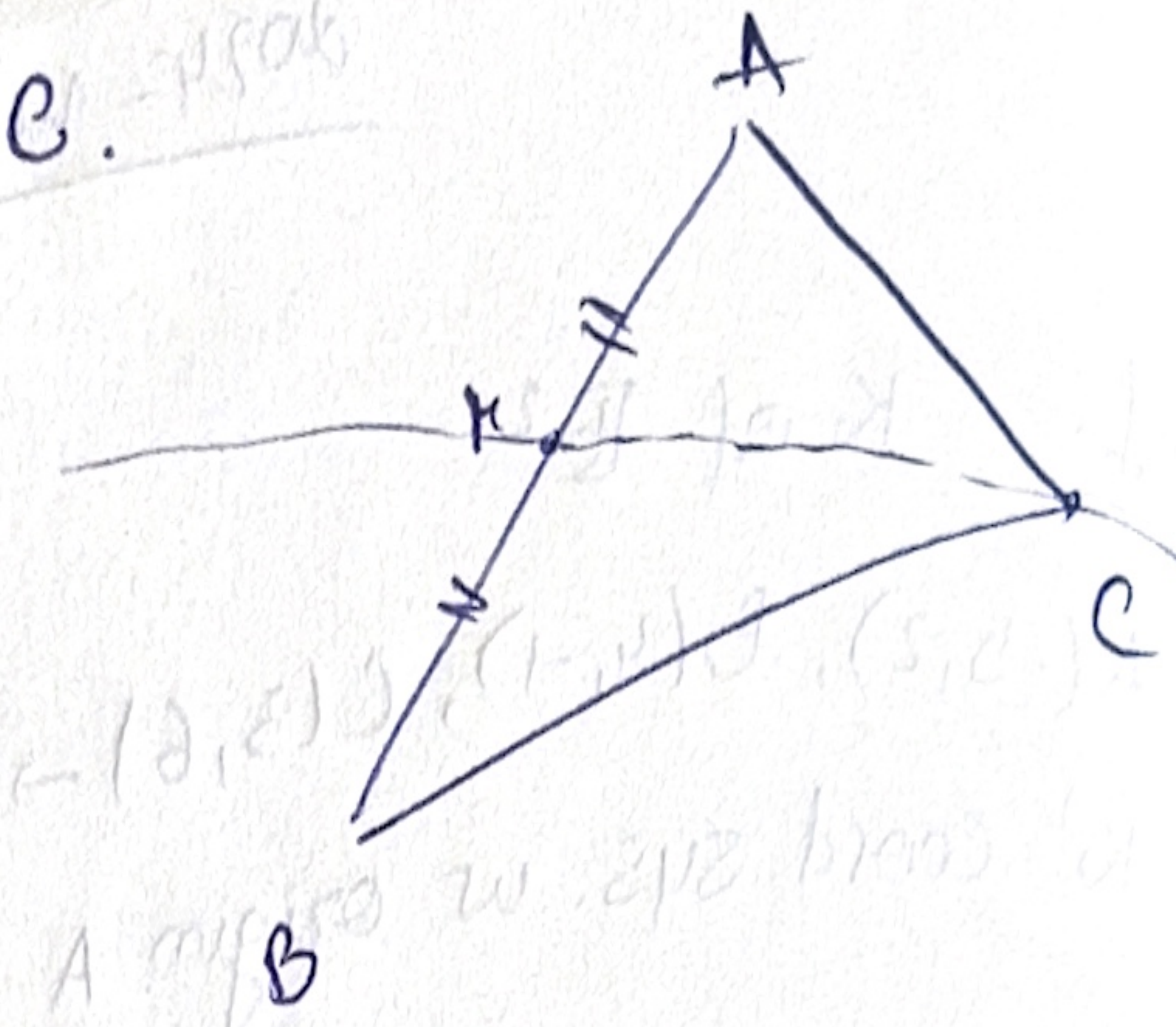
$$\Rightarrow \begin{cases} 3x - y - 3 = 0 \\ 2x + y - 7 = 0 \end{cases} \quad (+)$$

$5x - 10 = 0$

$x = 2$

$y = 3$ $\Rightarrow H(2,3)$

c) $K' \to$ eq. median ΔABC passing through C.



$A(-5,2),\ B(4,-1),\ C(3,6)$

$M\left(\frac{-5+4}{2}, \frac{2+(-1)}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$

$\vec{CM} = M - C = \left(-\frac{1}{2} - 3, \frac{1}{2} - 6\right) = \left(-\frac{7}{2}, -\frac{11}{2}\right)$

$\rightsquigarrow$ proportional direction vector: $(7,11)$.

- Line through $C(3,6)$ with dir. vector $(7,11)$: $\begin{cases} x = 3 + 7t \Rightarrow t = \frac{x-3}{7} \\ y = 6 + 11t \Rightarrow t = \frac{y-6}{11} \end{cases}$

$\Rightarrow \frac{x-3}{7} = \frac{y-6}{11} \Rightarrow 11(x-3) = 7(y-6) \Rightarrow 11x - 7y - 33 + 42 = 0 \Rightarrow$

$\Rightarrow 11x - 7y - 9 = 0.$

$$\boxed{A_\Delta = \frac{1}{2} \cdot \begin{vmatrix} x_B - x_A & y_B - y_A \\ x_C - x_A & y_C - y_A \end{vmatrix} = \frac{1}{2} \left| [\vec{AB}, \vec{AC}] \right|}$$

d) $A_{\Delta ABC} = ?$

$\vec{AB} = (9,-3),\ \vec{AC} = (8,4)$

$A_{\Delta ABC} = \frac{1}{2} \cdot \begin{vmatrix} 9 & -3 \\ 8 & 4 \end{vmatrix} = \frac{1}{2}(36 + 24) = \frac{1}{2} \cdot 60 = 30.$

$\vec{AB}$

$\vec{AC}$

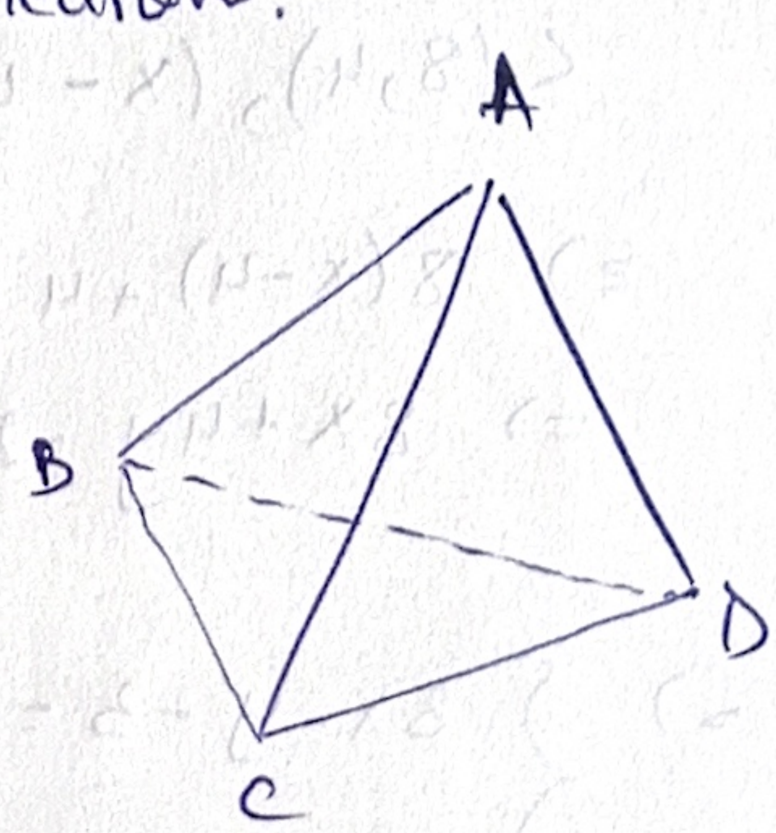
$$A_{\Delta ABC} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_C & y_C & 1 \end{vmatrix} = \begin{vmatrix} -5 & 2 & 1 \\ 4 & -1 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 5 + \cancel{24} \cancel{+6} - 8 - \cancel{30} + 3 =$$

$$\begin{matrix} 5 & 2 & 1 \\ 4 & -1 & 1 \end{matrix}$$

$= -3 + 3 = 0.\ \ ?!$

Pb2. $A(1,4,-1),\ B(1,5,1),\ C(-3,2,1),\ D(0,3,1)$ tetrahedron.

$K' : (C, \vec{AB}, \vec{AC}, \vec{AD})$



a) K - r - or.

$\vec{AB} = (0,1,2);\ \vec{AC} = (-4,-2,2),\ \vec{AD} = (-1,-1,2).$

$$M = \begin{bmatrix} 0 & -4 & -1 \\ 1 & -2 & -1 \\ 2 & 2 & 2 \end{bmatrix} \to \det M = \begin{vmatrix} \cancel{0} & -4 & -1 \\ 1 & \cancel{-2} & \cancel{-1} \\ \cancel{0} & 6 & 4 \end{vmatrix} =$$

$\ \ \downarrow\ \ \ \downarrow\ \ \ \downarrow$

$\ \ \vec{AB}\ \ \vec{AC}\ \ \vec{AD}$

$= 1 \cdot (-1)^3 \cdot \begin{vmatrix} -4 & -1 \\ 6 & 4 \end{vmatrix} = (-1) \cdot (-16 + 6) = (-1) \cdot (-10) = 10 > 0$

$\Rightarrow$ right-oriented.

b) eq. of plane π which contains [AB] and which is $\parallel$ to DC.

$\vec{AB} = (0,1,2) = u$. direction vector.

$\parallel DC \Rightarrow \vec{DC} = (3,1,0) = v$ direction.

$m = u \times v = (0,1,2) \times (-3,-1,0) =$
(normal vector) → the vector ⊥ to both u and v.
(cross product)

$$(0,1,2)\times(-3,-1,0) = \begin{vmatrix} i & j & k \\ 0 & 1 & 2 \\ -3 & -1 & 0 \end{vmatrix} = i \cdot \begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} + j \cdot \begin{vmatrix} 0 & 2 \\ -3 & 0 \end{vmatrix} + k \cdot \begin{vmatrix} 0 & 1 \\ -3 & -1 \end{vmatrix} =$$

$$= i \cdot (0+2) - j(0+6) + k \cdot (0+3) = 2i - 6j + 3k. \Rightarrow (2,-6,3)$$

$A(1,4,-1)$:

(plane eq:) $\pi: \; 2(x - x_A) - 6(y - y_A) + 3(z - z_A) = 0$

$\pi: \; 2(x-1) - 6(y-4) + 3(z+1) = 0$

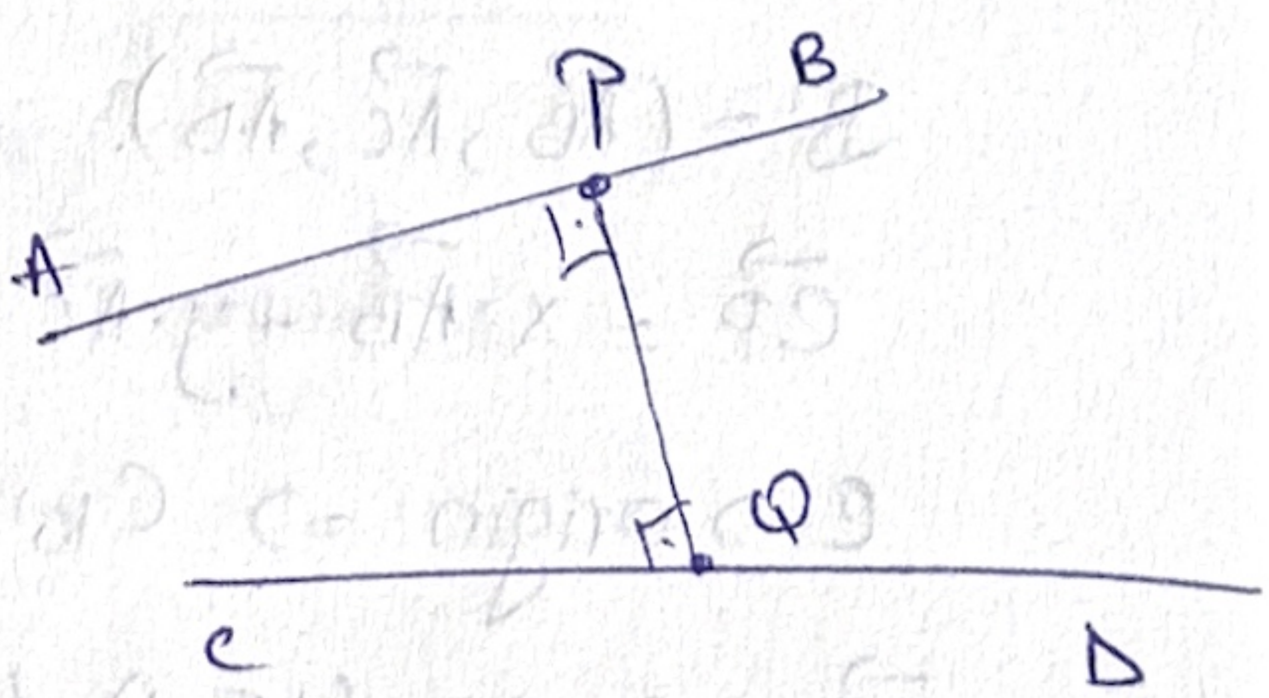
$\pi: \; 2x - 6y + 3z - 2 + 24 + 3 = 0$

$\pi: \; 2x - 6y + 3z + 25 = 0.$

c) common ⊥ of AB, CD.

(1) $l_{AB}: (x,y,z) = (\overset{A}{1,4,-1}) + s \cdot (0,1,2)$ ← lines in parametric form.

(2) $l_{CD}: (x,y,z) = (\overset{C}{-3,2,1}) + t \cdot (3,1,0)$



Let $P \in l_{AB}$, $Q \in l_{CD}$ $\Rightarrow$ PQ common ⊥ $\Leftrightarrow$

$$\begin{cases} PQ \perp AB \Rightarrow \vec{PQ} \cdot \vec{AB} = 0 \\ PQ \perp CD \Rightarrow \vec{PQ} \cdot \vec{CD} = 0. \end{cases}$$

(1): $(x,y,z) = (1, 4+s, -1+2s)$

(2): $(x,y,z) = (-3+3t, 2+t, 1)$

$$\vec{PQ} = Q - P = (-3+3t-1, 2+t-4-s, 1+1-2s) = (3t-4, t-s-2, 2-2s).$$

$\underline{\vec{PQ} \cdot \vec{AB} = 0} \Rightarrow (3t-4, t-s-2, 2-2s)(0,1,2) = 0$

$\Rightarrow 0(3t-4) + 1 \cdot (t-s-2) + 2(2-2s) = 0$

$\Rightarrow t - s - 2 + 4 - 4s = 0$

$\Rightarrow t - 5s + 2 = 0$

$\Rightarrow \underline{t = 5s - 2.}$ (1)

$\cancel{(3t-4)}$ $(3t-4, t-s-2, 2-2s)(3,1,0) = 0$

$\Rightarrow 3(3t-4) + 1(t-s-2) = 0$

$\Rightarrow 10t - 12 - s - 2 = 0$

$\Rightarrow \underline{10t - s - 14 = 0.}$ (2)

$\Rightarrow 10(5s-2) - s - 14 = 0$

$\Rightarrow 49s - 34 = 0 \Rightarrow s = \frac{34}{49}.$

$t = 5\cdot\frac{34}{49} - 2 = \frac{72}{49}$

$P = A + s\cdot(0,1,2) = \ldots = \left(1, \frac{230}{49}, \frac{19}{49}\right)$

$Q = C + t\cdot(3,1,0) = \ldots = \left(\frac{69}{49}, \frac{170}{49}, 1\right).$

$\vec{PQ} = Q - P = \left(\frac{69}{49} - 1, \frac{170-230}{49}, \frac{30}{49}\right) = \frac{10}{49}(2,-6,3)$
(direction)

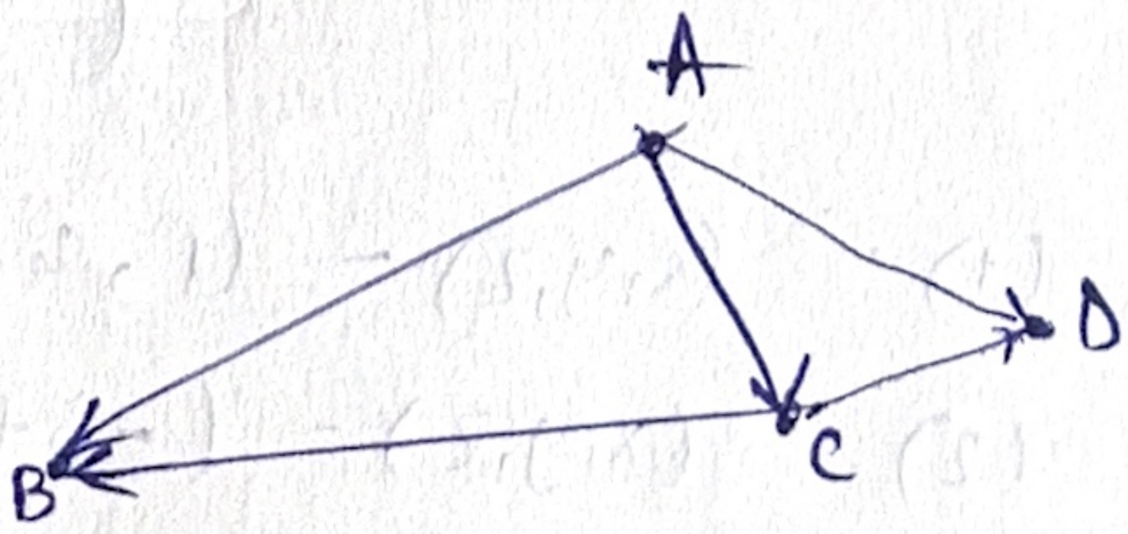
$$(x,y,z) = \underbrace{\left(1, \frac{230}{49}, \frac{19}{49}\right)}_{P} + \lambda\cdot\underbrace{(2,-6,3)}_{\vec{PQ}}.$$

d) $K' \to$ eq. of plane containing face (BCD) of tetrahedron.

$B' = (\vec{AB}, \vec{AC}, \vec{AD}).$

$\vec{CP} = x\cdot\vec{AB} + y\cdot\vec{AC} + z\cdot\vec{AD}$; $P(x,y,z)$

$C \to$ origin $\Rightarrow C_{K'} = (0,0,0)$



• $\vec{CB} = B - C = (1,5,1) - (-3,2,1) = (4,3,0).$

$\vec{AB} = (0,1,2)$
$\vec{AC} = (-4,-2,2)$
$\vec{AD} = (-1,-1,2)$
$\underline{\underline{\vec{CB}}} = \underline{\underline{\vec{AB}}} - \underline{\vec{AC}} = (4,3,0)$

$\Rightarrow \vec{CB} = x\cdot\vec{AB} + y\cdot\vec{AC} + z\cdot\vec{AD} = (4,3,0)$ $\Bigg\} \Rightarrow \begin{cases} x = 1, \\ y = -1 \\ z = 0 \end{cases} \Rightarrow B_{K'} = (1,-1,0)$

• $\vec{CD} = D - C = (3,1,0)$

$\underline{\underline{\vec{CD}}} = \underline{\underline{\vec{AD}}} - \underline{\vec{AC}} \Rightarrow \cancel{z} = 1, y = -1, x = 0 \Rightarrow D_{K'} = (0,-1,1).$

(BCD) contains: $C_{K'} = (0,0,0)$, $B_{K'} = (1,-1,0)$, $D_{K'} = (0,-1,1)$.

(BCD): $ax + by + cz = 0$

$B_{K'} \nearrow \Rightarrow a \cdot 1 + b \cdot (-1) + 0 \cdot c = 0 \Rightarrow a - b = 0 \Rightarrow a = b.$

$D_{K'} \nearrow \Rightarrow a \cdot 0 + b \cdot (-1) + c \cdot 1 = 0 \Rightarrow -b + c = 0 \Rightarrow b = c.$

$C_{K'} \nearrow \Rightarrow 0.$

$\Rightarrow a = b = c \Rightarrow$ ex: $a = b = c = 1 \Rightarrow x + y + z = 0.$

Pb3. Prove Jacobi: $\underline{(a \times b) \times c + (b \times c) \times a + (c \times a) \times b = 0.}$ (1)

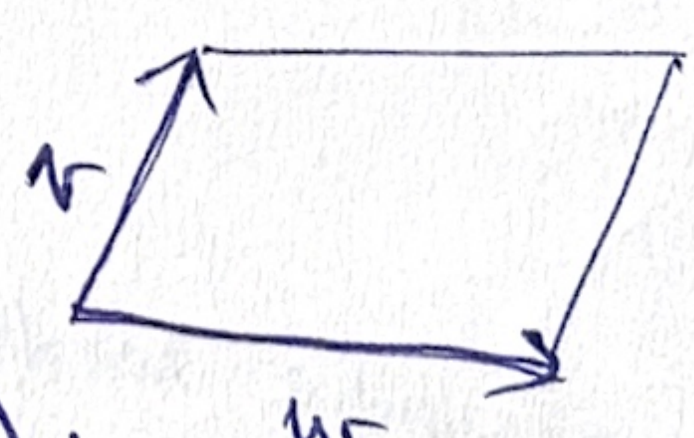
$(x \times y) \times z = \langle x; z \rangle y - \langle y, z \rangle x$

$$\begin{aligned}
(a \times b) \times c &= \cancel{\langle a, c \rangle b} - \cancel{\langle b, c \rangle a} \\
(b \times c) \times a &= \cancel{\langle b, a \rangle c} - \cancel{\langle c, a \rangle b} \\
(c \times a) \times b &= \cancel{\langle c, b \rangle a} - \cancel{\langle a, b \rangle c}
\end{aligned} \quad (+)$$

$\cdot (1) = 0.$

Pb4. (A parallelogram) in $\mathbb{E}^2$ in a Gram matrix.

$$\boxed{\text{Area}(P)^2 = \det G}$$



$$\underset{\text{Gram matrix}}{G} = \begin{pmatrix} \langle v, v \rangle & \langle v, w \rangle \\ \langle w, v \rangle & \langle w, w \rangle \end{pmatrix} = \begin{pmatrix} |v|^2 & \langle v, w \rangle \\ \langle v, w \rangle & |w|^2 \end{pmatrix}.$$

$\text{Area}(P)^2 = |v|^2 \cdot |w|^2 - \langle v, w \rangle^2$

If $v(v_x, v_y)$, $w(w_x, w_y)$ in an orthonormal frame

$$\Rightarrow \text{Area}(P) = \left| \det \begin{pmatrix} v_x & v_y \\ w_x & w_y \end{pmatrix} \right|$$

$$\Rightarrow \text{Area}(P)^2 = \det \begin{pmatrix} v_x & v_y \\ w_x & w_y \end{pmatrix}^2 = (v_x \cdot w_y - w_x \cdot v_y)^2 =$$

$$= v_x^2 \cdot w_y^2 - 2 v_x \cdot w_y \cdot w_x \cdot v_y + w_x^2 \cdot v_y^2$$

$$(v_x^2 + v_y^2)(w_x^2 + w_y^2) - (v_x \cdot w_x + v_y \cdot w_y)^2 =$$

$$= v_x^2 \cdot w_x^2 + \cancel{v_x^2 \cdot w_y^2} + v_y^2 \cdot w_x^2 + \cancel{v_y^2 \cdot w_y^2} - (\cancel{v_x^2 \cdot w_y^2} + 2 v_x w_x v_y w_y + \cancel{v_y^2 w_y^2})$$

$$= v_x^2 \cdot w_x^2 - 2 v_x w_x v_y w_y + v_y^2 \cdot w_x^2$$

$$= (v_x w_y - v_y \cdot w_x)^2$$

$$\text{Area}(\Phi)^2 = \underbrace{(v_x^2 + v_y^2)}_{\langle v, v \rangle} \underbrace{(w_x^2 + w_y^2)}_{\langle w, w \rangle} - \underbrace{(v_x w_x + v_y w_y)^2}_{\langle v, w \rangle^2}$$

$$= \langle v, v \rangle \cdot \langle w, w \rangle - \langle v, w \rangle^2 = \det G.$$

$$G = \begin{pmatrix} \langle v, v \rangle & \langle v, w \rangle \\ \langle w, v \rangle & \langle w, w \rangle \end{pmatrix}$$

~~→ Area~~